

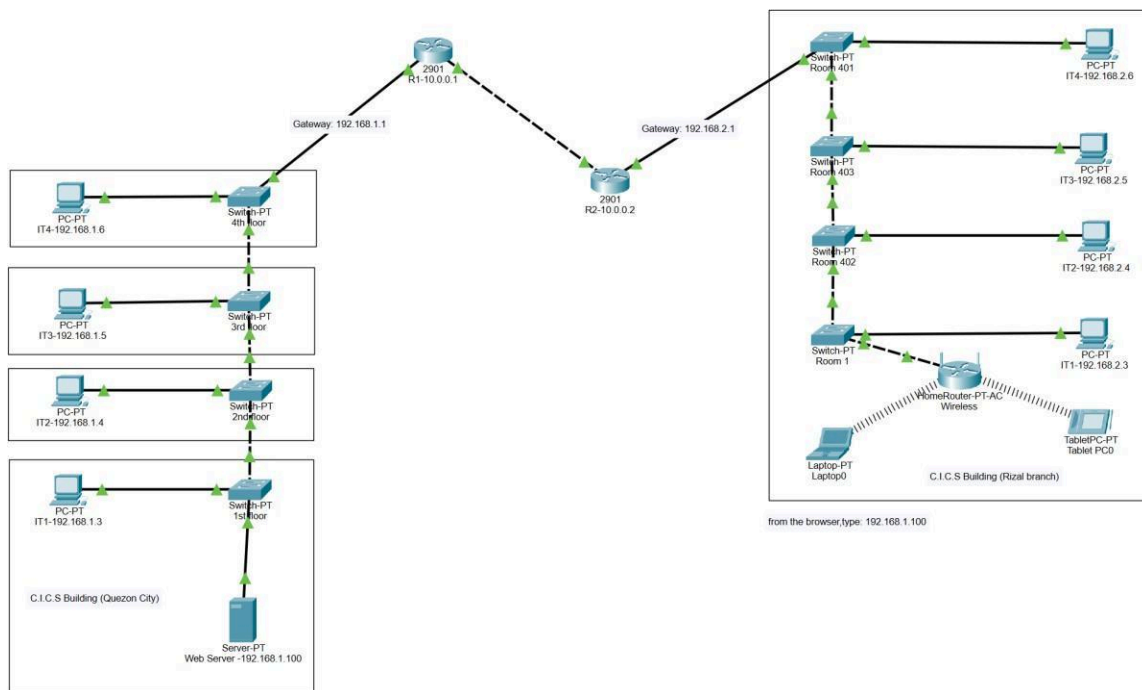
Network Management Lab Exercise 4.0: Wi-Fi Router Configuration and Web Server Setup

Objectives:

- Configure Cisco WiFi router and web server.
- Implement DHCP server from the Cisco WiFi router.
- Verify connectivity between web server (IP Address) and laptop or tablet connected in wifi router.
- Troubleshoot routing issues.

Topology:

Image of network topology with routers, switches, computers, laptop, tablet, and web server.



Tasks:

Initial Configuration:

- Assign IP addresses and subnet masks to router interfaces according to the provided topology.
- Configure default gateways on each computer according to their connecting router.
- Verify basic connectivity within each network segment using Ping.

Route Configuration:

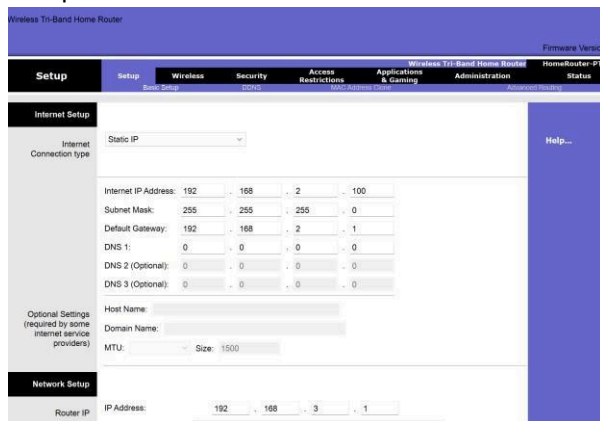
- On each router, configure dynamic routing protocols to reach all other networks. Use RIP.

Web Server configuration:

- Configure the IP Address and default gateways according to their connection in the diagram.
- Ensure that the HTTP services is enabled. Edit the index.html file and put the name of your members.

CISCO Wi-Fi Router configuration:

- Connect the cisco Wi-Fi router according to the diagram.
- Configure the static IP of the router in the Graphics User Interface (GUI) panel.
- Setup the DHCP service from the CISCO router for the laptop and tablet.



Testing and Verification:

- Use Ping and Traceroute to test connectivity between all computers across different network segments.
- Analyze the results and verify if packets are following the configured static routes.
- Open the Web browser on your laptop to access the website homepage. Ensure that you change the laptop adapter from a network card to a Wi-Fi card. Type the IP address of the webserver in the URL address bar.



- Access the website from the other computers.
- Use troubleshooting tools and techniques to identify and fix the issues. Analyze routing tables and debug logs to understand the problem and solution.

Reporting:

1. Document the configuration changes made throughout the lab.
2. Screenshot the testing you made using the Ping and Web browser tool.
3. Analyze the results of your tests and troubleshooting efforts.
4. Discuss the learnings and key takeaways from the lab exercise.